Vacuum switch

MP20/MP50 high-precision pressure switch with digital display

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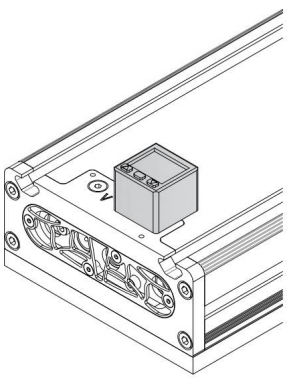
General -purpose



MP20/MP50 vacuum switch

Introduction

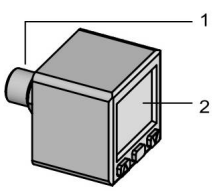
- Accuracy: Level 1
- 2-digital switching output, NPN/PNP can be switched.
- Small design and easy installation.
- "Current Value" and "Reference Value" are displayed at the same time, and the "reference value" can be set directly.
- Dual-screen, 3-color display (red, green and orange)
- Digital display, easier to identify



Application of MP20/MP50 vacuum switch

Application

- Use in vacuum and pressure system for process monitoring and optimizing cycle times.
- Adjusts the system loop and improves the economical efficiency of whole vacuum system.
- Cubic design is more suitable for panels installation.
- Applicable to all automatic handling system



Structure of MP20/MP50 vacuum switch

Structure

- (1) Connecting thread
- (2) Display

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MP20/MP50 vacuum switch purchase guide

MP20 - 000 - C1
① ② ③

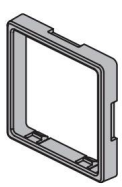
① Product series	② Output type	③ Installation fittings
MP20 - Compound pressure (pressure range: -100.0–100.0kPa) **MP50** - Positive pressure (pressure range: -0.100–1.000MPa) Note:the factory default value is-50Kpa;	000 - 2-digital switch value (NPN/PNP can be switched,factory default NPN) 050 - 1-digital switch value (NPN/PNP can be switched,factory default NPN)	Null - without mounting fittings C1 - with sensor mounting bracket (MS-FP-04) C2 - with sensor mounting bracket (MS-FP-05) D1 - with panel adapter (MS-FP-01/02) and frontal protective cover (MS-FP-03)

Model specifications of MP20/MP50 vacuum switch

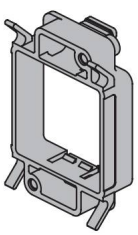
Model	Installation fittings		
	C1 (MS-FP-04)	**C2** (MS-FP-05)	**D1** (MS-FP-01/02+MS-FP-03)
MP20...	●	●	●
MP50...	●	●	●

Note:D1 means 01 type panel adapter(MS-FP-01)+ 02 type panel adapter(MS-FP-02)+03 type frontal protective cover

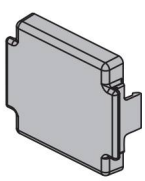
Accessories model specifications of MP20/MP50 vacuum switch



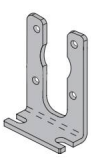
01 type panel adapter
(MS-FP-01)



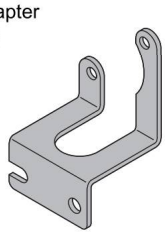
02 type panel adapter
(MS-FP-02)



03 type frontal protective cover
(MS-FP-03)



04 type sensor mounting bracket
(MS-FP-04)



05 type sensor mounting bracket
(MS-FP-05)

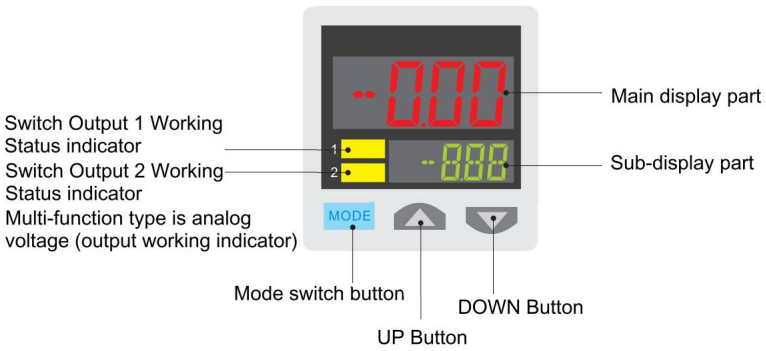
Name	Model
01 type panel adapter	MS-FP-01
02 type panel adapter	MS-FP-02
03 type frontal protective cover	MS-FP-03
04 type sensor mounting bracket	MS-FP-04
05 type sensor mounting bracket	MS-FP-05

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Description of MP20/MP50 vacuum switch display panel



Technical parameters of MP20/MP50 vacuum switch

Model		Positive pressure (MP50)	Compound pressure (MP20) [note 3]
Pressure		Standard pressure	
Rated pressure range [note 1]		-0.1MPa~1.0MPa	-100kPa~100kPa
Set pressure range [note 1]		-0.1MPa~1.0MPa	-100kPa~100kPa
Withstanding pressure		1.5MPa	500kPa
Applicable fluid		Non-corrosive gas	
Power supply voltage		12~24V DC±10%	
Current consumption		24V 30mA Max	12V 50mA Max
Switch output [note 2]		NPN O.C output: 80mA/24VDC Max or PNP O.C output: 80mA/24VDC	
Response time		±0.2% F.S.	±0.5% F.S.
Repetition precision		Selected by key operation :2.5ms,5ms,10ms,25ms,50ms,100ms,250ms,500ms,1000ms,5000ms	
analog output	Voltage output	0.6V~5V (or slightly smaller)	1V~5V (or slightly smaller)
		Linearity:±1%F.S. Output impedance: 1KΩ	
	Current output	2.4mA~20mA (or slightly smaller)	4mA~20mA (or slightly smaller)
		Linearity:Maximum load impedance when ±1%F.S. , the current output is 3000 when the power supply voltage is 12V, the current output is 6000 when the power supply voltage is 24V , and the minimum load impedance is 500 .	
The range of temperature		working temperature:0~50°C	storage temperature: -10-60°C(Non-condensation)
Humidity range		35~85%RH	
Specific temperatures		±1%F.S. (25°C)	
Protection level		IP40	
Material		Shell: nylon+glass fiber LCD: Propylene Pressure port: external thread POM+ internal M5 copper nut . Sealing ring: H-NBR Button: silicone rubber	
Size		30X30X25mm(Plastic parts) / 30X30X43mm(including connectors)	
Weight		about 80g (air gauge+user interface connector)	

[Note 1]: Due to the influence of temperature and linear compensation, there may be slight fluctuation near the upper/lower range of air gauge, which is normal.
(Note 2]: factory default NPN output
[Note 3]: factory default value is -50kPa.

Vacuum switch

MP20/MP50 high-precision pressure switch with digital display

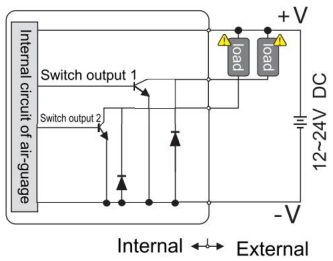
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Circuit diagram and wiring diagram

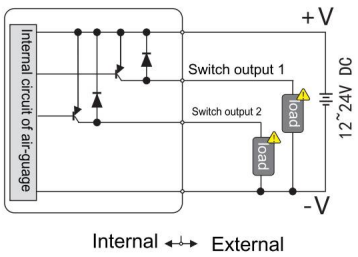
MP20/50-000 (2-digital output)

Please note that the load must be connected in series before use, otherwise the air gauge will be burned out.

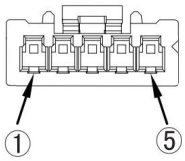
NPN output type



PNP output type



Terminal arrangement diagram

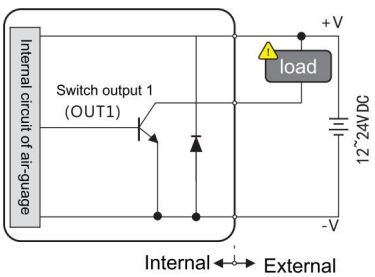


Pin of connector NO.	Terminal name
① Brown	+V
② Black	Switch output 1
③ White	Switch output 2
④ Orange	–
⑤ Blue	0V

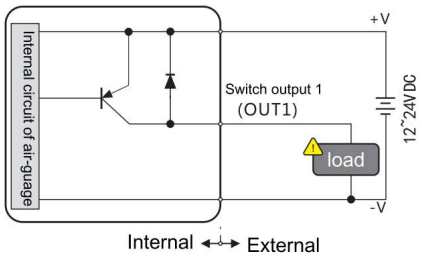
MP20/50-050/051 (1-digital output)

Please note that the load must be connected in series before use, otherwise the air gauge will be burned out.

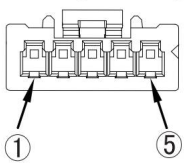
NPN output type



PNP output type



Terminal arrangement diagram

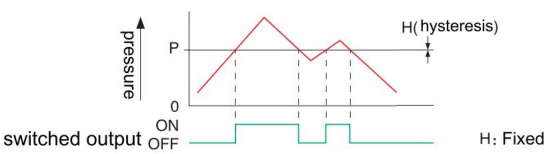


Pin of connector NO.	Terminal name
① Brown	+V
② Black	Switch output 1
③ White	–
④ Orange	–
⑤ Blue	0V

Input mode and output mode

① Easy mode

Mode for ON/OFF control of switch output.

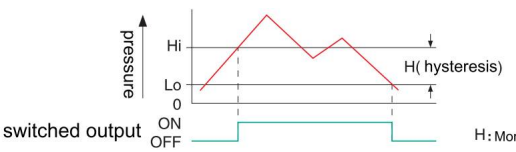


H: Fixed

Note: the hysteresis can be fixed on 8 level;

② Hysteresis mode:

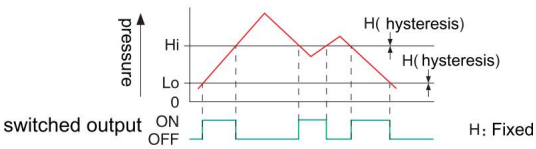
Arbitrarily set the hysteresis (lag) of switch output and the mode of ON/OFF control.



H: More than one position

③ Window comparator mode:

According to the pressure in the set range, the switch output is in the mode of ON or OFF.



H: Fixed

Note: the hysteresis can be fixed on 8 level;

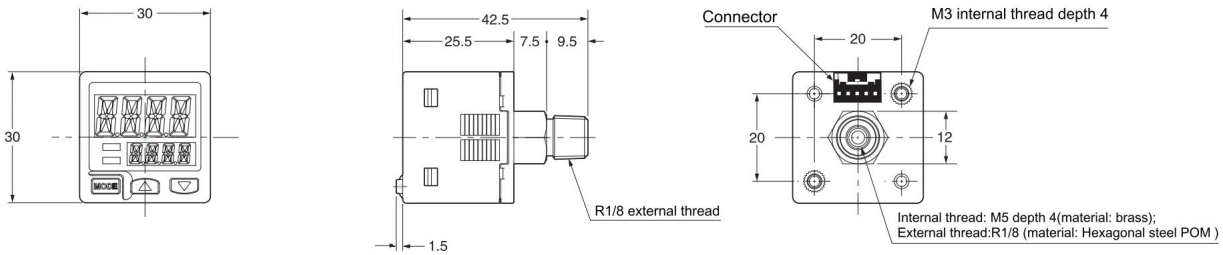
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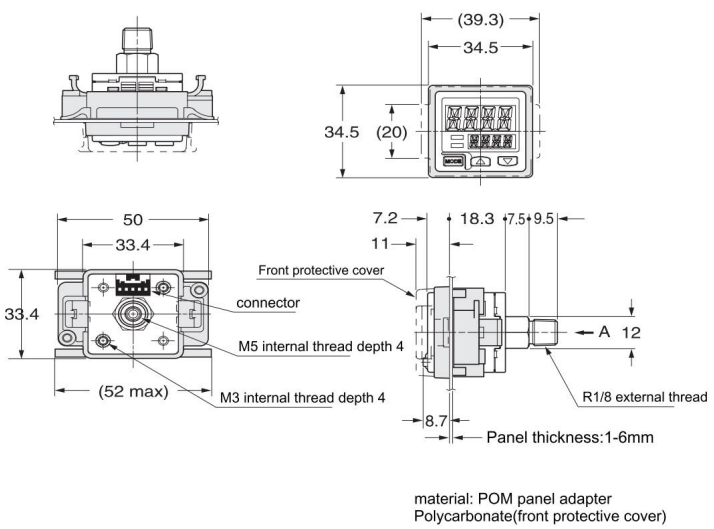
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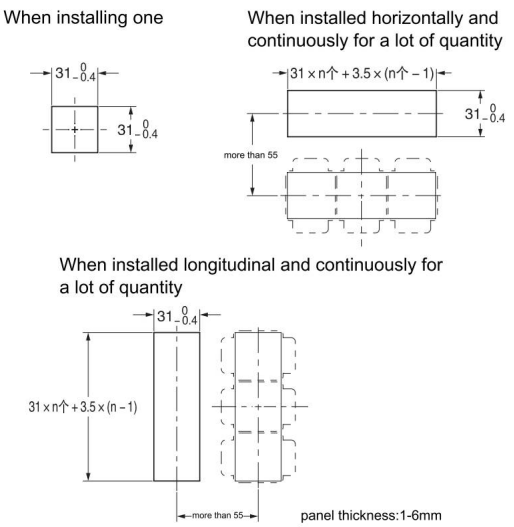
Design parameters of MP20/MP50 vacuum switch

MP20/MP50 vacuum switch...



MS-FP-01、MS-FP-02 Panel adapter、MS-FP-03 Front protective cover





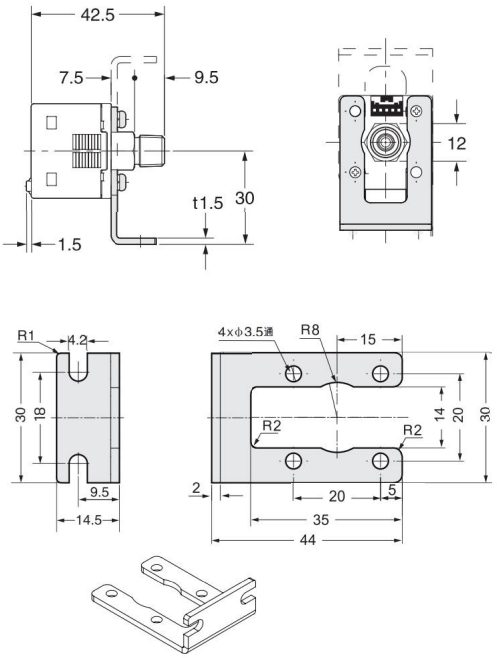
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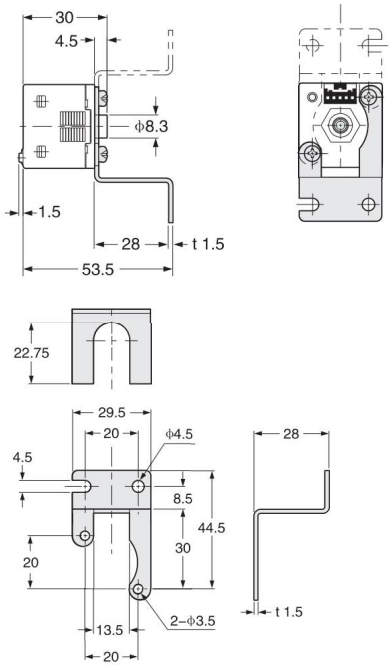
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Design parameters of MP20/MP50 vacuum switch

MS-FP-04 Sensor mounting bracket

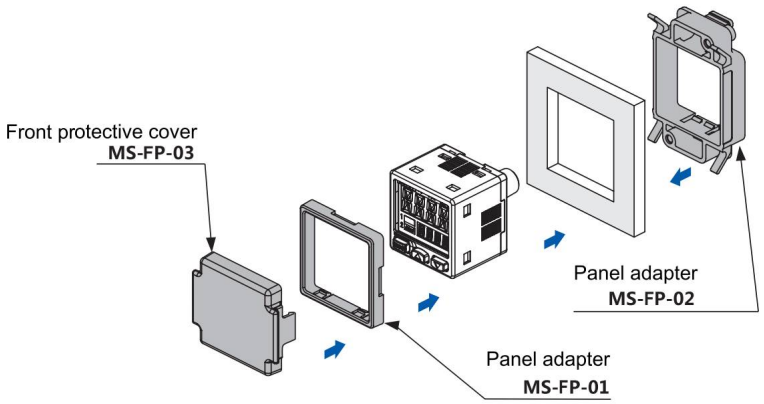


MS-FP-05 Sensor mounting bracket

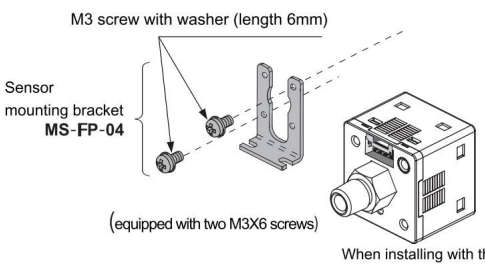


Installation instructions of vacuum switch MP20 /MP50

Installation instructions of panel adpter MS-FP-01/MS-FP-02, front protective cover MS-FP-03

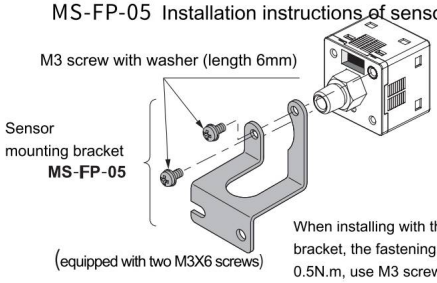


MS-FP-04 Installation instructions of sensor mounting bracket



When installing with the matching mounting bracket, the fastening torque should be less than 0.5N.m, use M3 screw and the length is 6mm.

MS-FP-05 Installation instructions of sensor mounting bracket



When installing with the matching mounting bracket, the fastening torque should be less than 0.5N.m, use M3 screw and the length is 6mm.